Anil Koushik N P

Specialist Scientist · Centre for Infrastructure, Sustainable Transportation and Urban Planning (CiSTUP), Indian Institute of Science, Bengaluru

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Research Profile

Transportation researcher working on data-driven travel demand modeling and urban mobility. My doctoral work developed machine learning methods for activity-based travel demand models, including LSTM-based activity schedule models, transfer learning for spatial transferability, and SHAP-based interpretation of neural network models. At IISc, I develop trip-based and activity-based travel demand models for Bengaluru and support the software tools built on them.

Research interests: activity-based travel demand modeling; machine learning for travel behavior; interpretable AI; spatial transferability; choice modeling; data-driven urban mobility planning.

Research and Professional Experience

Specialist Scientist

CiSTUP, Indian Institute of Science, Bengaluru | Apr 2025 – present

- Develop trip-based and activity-based travel demand models for the city-scale Bengaluru travel demand model.
- Provide technical guidance to the software team building travel demand modeling tools, and test and validate the tools.
- Supervise household and transit surveys.
- Mentor Master's and PhD students on their theses.
- Contribute to early-stage research on inferring land use from satellite imagery, GPS trajectories, transit ticketing and points-of-interest data, and on on-demand shared taxi and auto-rickshaw services for metro station access.

Research Scientist

Foundation for Science Innovation and Development (FSID) and CiSTUP, IISc, Bengaluru | Nov 2023 – Apr 2025

- Developed trip-based and activity-based models for the Bengaluru travel demand model; supported tool development, testing and household and transit surveys.
- Mentored Master's students on their theses.

Assistant Professor, Department of Civil Engineering

PES University, Bengaluru | Jul 2015 – Dec 2017

- Taught Transportation Engineering, Highway Engineering, Railway and Airport Engineering, Traffic Engineering and Engineering Mechanics.

EPMA Application Support Engineer

Advanced Facility for Microscopy and Microanalysis, IISc | May 2014 – Jun 2015

- Supported chemical analysis of samples using electron probe microanalysis.

Senior Research Fellow

Indian Institute of Science | Apr 2013 – Apr 2014

- Petrographic, chemical and computational study of concrete at high temperatures.

Senior Design Engineer

Larsen & Toubro, ECC Division | Jul 2008 – Jul 2010

- Designed temporary works for the Mumbai Monorail project and the launching girder for the Chennai Metro Rail project.

Education

Ph.D., Transportation Engineering

Indian Institute of Technology Delhi | Jan 2018 – Aug 2024

Thesis: *Data-driven methods for activity schedule modeling*

- Represented the daily activity schedule as a categorical time series and modeled it with LSTM networks, capturing all schedule dimensions jointly and satisfying the time-budget constraint by construction.
- Assessed and improved the spatial transferability of the models using transfer learning.
- Interpreted the models using SHapley Additive exPlanations (SHAP).
- Proposed an ANN framework with correlated error terms for activity pattern modeling, allowing flexible substitution patterns.

M.Tech., Transportation and Infrastructure Engineering

Indian Institute of Science, Bengaluru · CGPA 6.7/8 · GATE qualified | 2010 – 2012

- Project: non-linear finite element analysis of post-tensioned concrete girders, accounting for multiaxial stress effects on concrete failure.

B.Tech., Civil Engineering

National Institute of Technology Karnataka, Surathkal · CGPA 6.8/10 · AIEEE qualified | 2004 – 2008

Journal Publications

J1. **Koushik, A.**, Manoj, M., & Nezamuddin, N. (in press). Artificial neural networks with correlated error terms for travel behavior modeling. *Transportation Research Record*

J2. Feroz, M. N., Nirmale, S. K., **Koushik, A.**, & Pinjari, A. R. (2026). Modelling multimodal mode choice behaviour considering spatial variability in level of service attributes. *Transportation Research Part A: Policy and Practice, 204, 104710*. doi:10.1016/j.tra.2025.104710

J3. Kachroo, P., **Koushik, A.**, & Manoj, M. (2024). Automatic activity-travel sequence generator using language, grammar, and machine theory. *Transportation Letters, 1–10*. doi:10.1080/19427867.2024.2416309

J4. **Koushik, A.**, Manoj, M., & Nezamuddin, N. (2024). SHapley Additive exPlanations for explaining artificial neural network based activity schedule models. *Transportation Letters*. doi:10.1080/19427867.2024.2359304

J5. **Koushik, A.**, Manoj, M., & Nezamuddin, N. (2024). SHapley Additive exPlanations for explaining artificial neural network based mode choice models. *Transportation in Developing Economies, 10, 12*. doi:10.1007/s40890-024-00200-6

J6. **Koushik, A.**, Manoj, M., Nezamuddin, N., & Prathosh, A. P. (2023). Activity schedule modeling using machine learning. *Transportation Research Record, 1–23*. doi:10.1177/03611981231155426

J7. **Koushik, A. N.**, Manoj, M., Nezamuddin, N., & Prathosh, A. P. (2022). Testing and enhancing spatial transferability of artificial neural networks based travel behavior models. *Transportation Letters, 1–12*. doi:10.1080/19427867.2022.2130150

J8. **Koushik, A. N. P.**, Manoj, M., & Nezamuddin, N. (2020). Machine learning applications in activity-travel behavior research: a review. *Transport Reviews, 40, 288–311*. doi:10.1080/01441647.2019.1704307

Conference Presentations

C1. **Koushik, A.**, Manoj, M., & Kodamana, H. (2024). Mode choice modeling using variational autoencoders. *International Choice Modelling Conference (ICMC), Puerto Varas, Chile*

C2. **Koushik, A.**, & Manoj, M. (2023). Detecting variable interactions in mode choice models using artificial neural networks. *7th Conference of the Transportation Research Group of India (CTRG), Surat*

C3. **Koushik, A.**, Manoj, M., & Nezamuddin, N. (2022). SHapley Additive exPlanations for explaining artificial neural network based mode choice models. *Transportation Planning and Implementation Methodologies for Developing Countries (TPMDC), IIT Bombay*

C4. **Koushik, A.**, Manoj, M., & Nezamuddin, N. (2022). Explaining artificial neural network-based mode choice models using SHapley Additive exPlanations. *4th Bridging Transportation Researchers Conference*

C5. **Koushik, A.** (2019). Activity schedule modeling using LSTM networks. *Autumn School on Big Data and Methods in Regional Science, Tata Institute of Social Sciences, Mumbai*

C6. **Koushik, A.**, Sangeeta, Suma, Sandhya, & Murthy, K. V. (2017). Testing the utility of hide glue in bituminous concrete. *ASCE India Conference, IIT Delhi*

Research Collaborations

Activity-travel sequence generation using language, grammar and machine theory
With Dr. Manoj M (IIT Delhi) and Prof. Pushkin Kachroo (University of Nevada, Las Vegas)
Treated activity sequences as strings and applied formal language, grammar and automata theory; developed a stochastic finite state machine that generates activity sequences matching observed frequencies.

Choice modeling using variational autoencoders
With Dr. Manoj M and Dr. Hariprasad Kodamana (IIT Delhi)
Developed a VAE with a Gaussian-mixture latent space whose encoder classifies individuals by socio-demographics and whose decoder reproduces the inputs while predicting choices from trip attributes.

Teaching and Invited Lectures

- Guest lectures on artificial neural networks, support vector machines and deep neural networks, and on text analysis in travel behavior modeling. Short-Term Training Program on Advanced Statistical and Machine Learning Techniques in Transportation and Logistics, Gati Shakti Vishwavidyalaya (2023).
- Teaching assistant, TEQIP short-term course on Machine Learning for Engineering Applications.
- Undergraduate courses in transportation and civil engineering, PES University (2015–2017).

Professional Service

- Reviewer, *Transport Reviews*
- Reviewer, Bridging Transportation Researchers Conference (2022)
- Thesis reviewer, Eric Pas Dissertation Prize (2020)

Awards and Fellowships

- MHRD Fellowship for doctoral studies, IIT Delhi (2018 – 2022)
- Contingency grant to present at TPMDC, Department of Civil Engineering, IIT Delhi (2022)
- MHRD Fellowship for M.Tech. studies, IISc (2011 – 2012)

Technical Skills

Programming	Python, R, GAUSS
Machine learning	ANN, LSTM, VAE, transfer learning, SHAP
Modeling	Trip-based and activity-based travel demand models, discrete choice models
Geospatial	QGIS

Workshops and Training

- Short course on Advanced Choice Modelling Methods with Application in Transportation and Urban Systems, IISc (2024)
- Autumn School on Big Data and Methods in Regional Science, TISS Mumbai (presented a paper) (2019)
- Author Workshop on How to Publish a Technical Paper, Wiley and Central Library, IIT Delhi (2018)
- Performance Modeling and Simulation, Centre for Continuing Education, IISc (2016)
- Workshop on Urban Accessibility and Mobility Improvements, IISc (2012)

References

Available on request.